YAKEEN

NEET 2.0 2027

Physics | Lecture - 04

Units and Measurement

By- Manish Raj (MR Sir)



Todays Goal

PhD on dimⁿ analysis.

Singnifit fig.

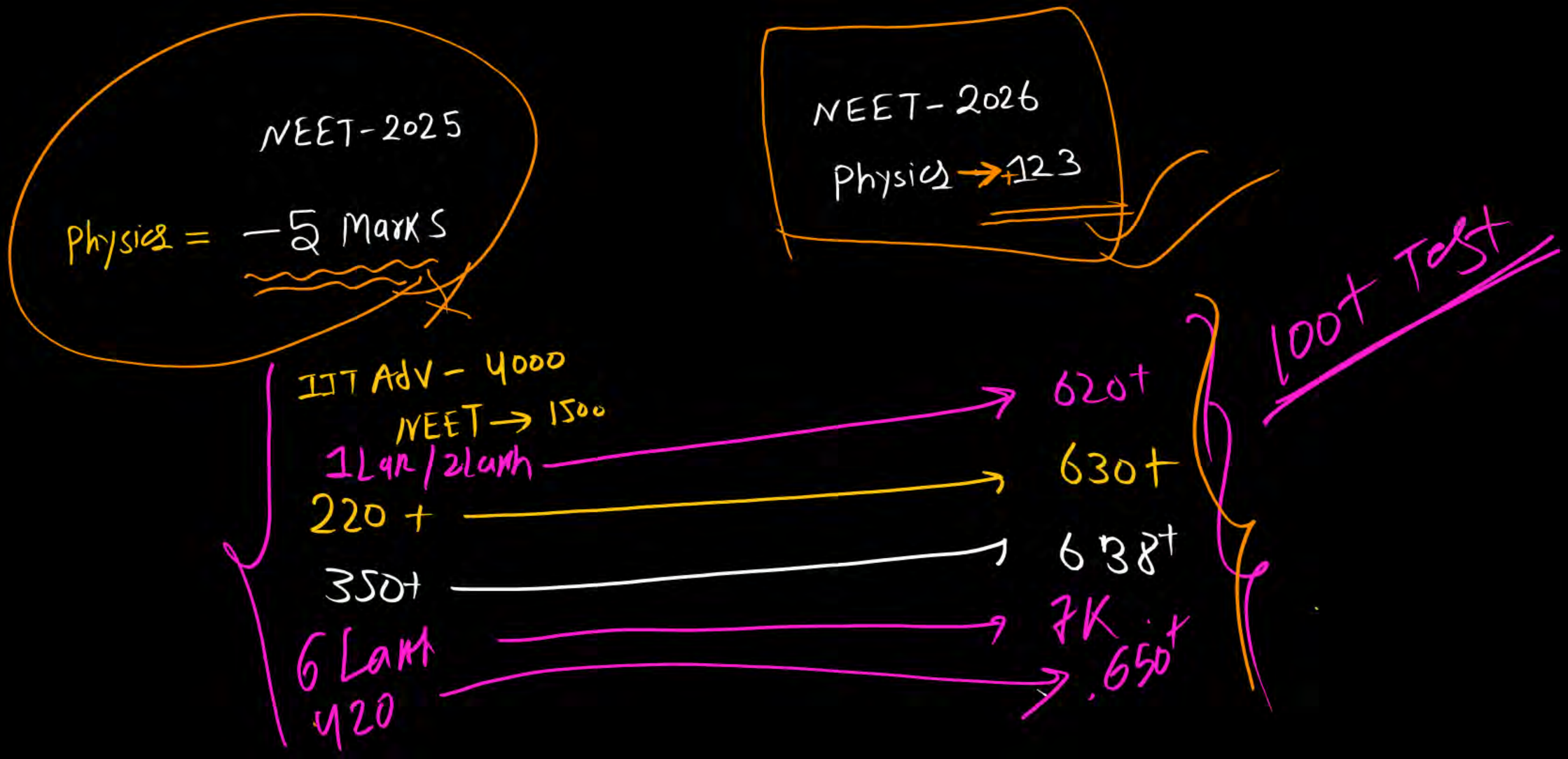
H/W

⇒ U.d.M Sangh assignnp 1 & 2 karlo.

→ Maha Manthu vector

⇒ M.R. Physics of Unit and dimenisin.. only do marked Question no.




NEET-2025
Physics = −5 Marks
NEET-2026
Physics → +123
IIT Adv - 4000
NEET → 1500
1 Lakh / 2 Lakh
220 +
350+
6 Lakh
420
620+
630+
638+
7K
650+
100+ Test

Question n/w only solve on rough Paper

(1) Velocity $V = \frac{\alpha}{\beta + \sqrt{\text{density}}}$, find dimension of α and β.

$V = \frac{\alpha}{\sqrt{\text{density}}}$ $\quad \beta = \sqrt{\text{density}}$

Question

(2)

Force $F = A \cos(Bx) + C \sin(Dt)$

Find dimension of $\left[\frac{AD}{B}\right]$

Sol:

$$\begin{cases} Bx = 1 \\ Dt = 1 \end{cases}$$

$$F = A = C$$

$$\frac{AD}{B} = \frac{Fx}{T} = \checkmark\checkmark$$

Question

(3)

Force $F = \sin\left(\frac{A}{t} + \frac{B}{x^2}\right)$

Find dimension of AB and A/B?

Sol:

$$\frac{A}{t} = \frac{B}{x^2}$$

$$\frac{A}{B} = \left(\frac{t}{x^2}\right)$$ ✓

Question

Pressure P = A $\log_e(B/t - CV)$

Find dimension of AB and BC?

Sol:

$P = A$

$\frac{B}{t} = CV = 1$

$\frac{B}{t} = 1$

$B = t$

$AB = P \times t$

$= ML^{-1}T^{-2} \times T$

$= ML^{-1}T^{-1}$

$BC = t \times \frac{1}{V}$

Question

Velocity of object $V = A\, e^{(Bx + C/t + D)}$.

Find dimension of AB and C/D?

Sol:

$V = A$

$Bx + \frac{C}{t} + D = 1$

$AB = V \times \frac{1}{x}$

$= \frac{V}{x} = \frac{LT^{-1}}{L} = T^{-1}$

$\left(\frac{C}{D}\right) = \frac{t}{1}$

Question

6

Energy of object $E = B\sin\{B(Ct - \sqrt{x}\,D + E)\}$

Find dimension of B, C and D?

dimless

B x0 and x2 ko tab Angle h.

Sol:

"$Ct = \sqrt{x}D = E$"

$Bct - B\sqrt{x}D + BE = Angle$

Energy $E = B \times 1$

$Bct = BD\sqrt{x} = BE = 1$

$E = ML^2T^{-2} = B$ B_2

$C = \frac{1}{Bt} = \frac{1}{ML^2T^{-2}T}$

$C = M^{-1}L^{-2}T^{2}$

Question

(a) JEE

Displacement of object $y = \frac{A\sqrt{x}}{B+\sqrt{t}}$ find dimension of A/B = ?

Sol:

$$Y = \frac{A\sqrt{x}}{B}$$

$$\frac{A}{B} = \frac{Y}{\sqrt{x}}$$

Question

If maximum acceleration of oscillating particle is α and maximum velocity is β, then find time period

[NEET 2013]

S.H.M

$a_{max} = \alpha$, β = velocity

1. $2\pi\frac{\alpha}{\beta}$

2. $2\pi\alpha\beta$

3. $2\pi\frac{\beta}{\alpha}$

4. $\frac{2\pi\alpha^2}{\beta^2}$

$a = \frac{v}{t}$

$t = \frac{v}{a} = \frac{\beta}{\alpha}$

Question

No copy

On the basis of dimension, decide which of the following relation for displacement of a particle is not correct [NCERT]

1. $y = a \sin\left(\frac{2\pi t}{T}\right)$ ← Angle = dimⁿ less.

2. $y = a \sin\left(\frac{vt}{\lambda}\right)$ ← dimⁿ less = $\left(\frac{vt}{\lambda}\right)$

3. $y = \sqrt{2}a \sin\left(\frac{2\pi t}{T}\right) \cos\left(\frac{2\pi x}{\lambda}\right)$

4. $y = \frac{x}{t} \sin(\lambda t)$ (Wrong)

Ans (4)

Angle = $\lambda t = LT$ ✗

must be dimⁿ less

Question

If force (F), acceleration (a) and time t is used as a fundamental P.Q. then find dimension of length in terms of them:

1. $F^0 a^1 T^2$
2. $F a^2 T^2$
3. $F a^2 T^0$
4. $F^0 a T$

$\rightarrow l = F^{\alpha} a^{\beta} t^{\gamma}$

MR*

$M^0 L^1 T^0 = (MLT^{-2})^{\alpha} (LT^{-2})^{\beta} (T)^{\gamma}$

$= F^0 \; a^1 T^2$

The MR* $l = \left(\frac{1}{2}\right) a t^2$

$= F^0 a^1 T^2$

Question

If force, acceleration and time are basic fundamental P.Q. then find dimension of energy.

1. $F^2 A^{-1} T$
2. $F A T^2$
3. $F A T^{-2}$
4. $F A^{-1} T$

The mR^x

$E = F \times l$

$= F' \times a \cdot t^2$

Question

Copy

If energy E, velocity V and time T are taken as fundamental units, the dimensional formula for surface tension is

1. $[E\ V^{-2}\ T^{-2}]$
2. $[E^{-2}\ V\ T^{-2}]$
3. $[E^{-2}\ V\ T^{-2}]$
4. $[E^{-2}\ V^{-2}\ T^{2}]$

class video में
Solve kar

Question

copy

If force (F), acceleration (a) and time t is used as a fundamental P.Q. then find dimension of length in terms of them:

1. $F^0 a^1 T^2$
2. $F a^2 T^2$
3. $F a^2 T^0$
4. $F^0 a T$

Class video MO
Solve hai

Question

Copy

$l = vt$

NEET

If energy E, velocity V and time T are taken as fundamental units, the dimensional formula for surface tension is

1. $[E\ V^{-2}\ T^{-2}]$
2. $[E^{-2}\ V\ T^{-2}]$
3. $[E^{-2}\ V\ T^{-2}]$
4. $[E^{-2}\ V^{-2}\ T^{2}]$

hint → $S = \frac{F}{l}$

$= \frac{MLT^{-2}}{L}$

$= MT^{-2}$

MR^2

$S = E^{\alpha} V^{\beta} T^{\gamma}$

$MT^{-2} = (ML^2T^{-2})^{\alpha} (LT^{-1})^{\beta} (T)^{\gamma}$

$= ML^{2\alpha} T^{-2\alpha} L^{\beta} T^{-\beta} T^{\gamma}$

$= E^1$

$\alpha = 1 \quad \beta = -2 \quad \gamma = -2$

The MR^2

$S = \frac{F \times l}{l \times l}$

$= \frac{E}{l^2} = \frac{E}{V^2 T^2}$

$S = E\ V^{-2}\ T^{-2}$

Question

No Copy

If force F, area A and density D are taken as the fundamental units, the representation of Young's modulus 'Y' will be:

1. $[F^{-1} A^{-1} D^{-1}]$
2. $[F A^{-2} D^{2}]$
3. $[F A^{-1} D]$
4. $[F A^{-1} D^{0}]$ ✓

→ same as pressure

$$Y = \frac{F}{A} = M L^{-1} T^{-2}$$

$$Y = \frac{F}{A} = F A^{-1} D^{0}$$

Question 1

Plank's constant (h), speed of light in vacuum (c) and Newton's gravitational constant (G) are three fundamental constants. Which of the following combinations of these has the dimension of length? **[NEET 2016]**

1. $\frac{\sqrt{hG}}{c^{3/2}}$

2. $\frac{\sqrt{hG}}{c^{5/2}}$

3. $\sqrt{\frac{hc}{G}}$

4. $\frac{\sqrt{Gc}}{h^{3/2}}$

$E = hf$

$\Rightarrow h = M^1 L^2 T^{-1}$

$\Rightarrow G = M^{-1} L^3 T^{-2}$

$\Rightarrow C = L T^{-1}$

$l = h^\alpha G^\beta C^\gamma$

$M^0 L^1 T^0 = \sqrt{hG} / c^{3/2}$

$= \sqrt{M^{-1} L^2 \times L^3} = \frac{(L^{5/2})}{L^{3/2}} = L^1$

MR* Box

(1) Sabse pahle mass ka dimn match karo.

Sirf mass ko dekho

Question

No copy

$F = 6\pi\eta r v_T$ $\quad \eta = \frac{F}{r v_T} = \frac{ML T^{-2}}{L \times L T^{-1}} = ML^{-1}T^{-1}$

If dimensions of critical velocity v_c of a liquid flowing through a tube are expressed as $[\eta^x \rho^y r^z]$ where η, ρ, r are the coefficient of viscosity of liquid, density of liquid and radius of the tube respectively, then the values of x, y and z are given by

AIPMT → (2014)

1. $1, 1, 1$ ✗
2. $1, -1, -1$ ✓ Answer
3. $-1, -1, 1$
4. $-1, -1, -1$ ✗

$v_c = LT^{-1}$

$V_c = \eta^x \rho^y r^z$

$(M^0 L T^{-1}) = (M^1 L^{-1} T^{-1})^x \left(\frac{M^2}{L^3}\right)^y (L)^z$

$= \frac{\eta}{\rho}$ or $\frac{\rho}{\eta}$ $\quad \frac{1}{\rho\eta}$ ✗



No copy

Question

$F = \frac{e^2}{4\pi\varepsilon_0 r^2}$

$\frac{e^2}{4\pi\varepsilon_0} = Fr^2 = [M L^3 T^{-2}]$

A physical quantity of the dimensions of length that can be formed out of c, G and $\frac{e^2}{4\pi\varepsilon_0}$ is [c is velocity of light, G is universal constant of gravitation and θ is charge]

1. $\frac{1}{c^2}\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$ → $c = LT^{-1}$, $G = M^{-1}L^3T^{-2}$

2. $c^2\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$

3. $\frac{1}{c^2}\left[\frac{e^2}{G4\pi\varepsilon_0}\right]^{1/2}$ → $\left[\frac{e^2}{4\pi\varepsilon_0}\right] = M^1L^3T^{-2}$

4. $\frac{1}{c}G\frac{e^2}{4\pi\varepsilon_0}$

$l = c^x G^y \left(\frac{e^2}{4\pi\varepsilon_0}\right)^z$

$M^0 L^1 T^0 = \left(G \times \left(\frac{e^2}{4\pi\varepsilon_0}\right)\right)^{1/2} \times z$

$= M^{-1} M^1 (L^3)^{1/2} = \frac{L^3}{L^2} = L$

$= M^0$

Question

$F = \frac{e^2}{4\pi\varepsilon_0 r^2}$

$\frac{e^2}{4\pi\varepsilon_0} = Fr^2 = [M L^3 T^{-2}]$

$F = \frac{Gm^2}{r^2}$

A physical quantity of the dimensions of length that can be formed out of c, G and $\frac{e^2}{4\pi\varepsilon_0}$ is [c is velocity of light, G is universal constant of gravitation and θ is charge]

1. $\frac{1}{c^2}\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$

$\rightarrow c = LT^{-1}$

$\rightarrow G = M^{-1}L^3T^{-2}$

$\rightarrow \left[\frac{e^2}{4\pi\varepsilon_0}\right] = M^1L^3T^{-2}$

2. $c^2\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$

3. $\frac{1}{c^2}\left[\frac{e^2}{G4\pi\varepsilon_0}\right]^{1/2}$

4. $\frac{1}{c}G\frac{e^2}{4\pi\varepsilon_0}$

$l = c^x G^y \left(\frac{e^2}{4\pi\varepsilon_0}\right)^z$

$z - y = 0 \quad -①$

$x + 3y + 3z = 1 \quad -②$

$-x - 2y - 3z = 0 \quad -④$

$M^0L^1T^0 = (LT^{-1})^x (M^{-1}L^3T^{-2})^y (M^1L^3T^{-2})^z$

$M^0L^1T^0 = M^{-y+z} L^{x+3y+3z} T^{-x-2y-3z}$

Question

Copy

In a new system of units energy (E), density (d) and power (P) are taken as fundamental units, then the dimensional formula of universal gravitational constant G will be

1. $[E^{-1} d^{-2} P^{2}]$
2. $[E^{-2} d^{-1} P^{2}]$
3. $[E^{2} d^{-1} P^{-1}]$
4. $[E^{1} d^{-2} P^{-2}]$

$E = ML^2T^{-2}$

$d = ML^{-3}$

$P = ML^2T^{-3}$

$G = M^{-1}L^3T^{-2}$

IIT-Advance

Time pe aftaur Karo

$G = E^x P^y d^z$

$M^{-1}L^3T^{-2} = (E)^x (P)^y (d)^z$

(a) $\frac{P^2}{E} = \frac{T^{-6}}{T^{-2}} = T^{-4}$

(b) $\frac{P^2}{E^2} = \frac{T^{-6}}{T^{-4}} = T^{-2}$

(c) $\frac{E^2}{P} = \frac{T^{-4}}{T^{-3}} = T^{-1}$ X

(d) $= \frac{E}{P^2}$ X

Question

Copy

The speed of light C, gravitational constant G and plant constant h are taken as fundamental P.Q then the dimension of time in the new system of unit

JEE-Adv

1. $G^{1/2} h^{1/2} c^{-5/2}$
2. $G^{-1/2} h^{1/2} c^{-1/2}$
3. $G^{-1/2} h^{1/2} c^{-3/2}$
4. $G^{-1/2} h^{1/2} c^{1/2}$

$t = c^x G^y h^z$

$M^0 L^0 T^1 = [G h / L^{5/2}]$

$C = LT^{-1}$

$G = M^{-1} L^3 T^{-2}$

$h = M L^2 T^{-1}$

$= [M^1 M^1 L^3 \times L^2] = \frac{L^{5/2}}{L^{5/2}} = L^0$ ✓

Question

H|W

The frequency of vibrations f of a mass m suspended from a spring of spring constant K is given by a relation of type $f = cm^x K^y$, where c is a dimensionless constant. The values of x and y are:

1. $x = \frac{1}{2}, y = \frac{1}{2}$
2. $x = \frac{-1}{2}, y = \frac{-1}{2}$
3. $x = \frac{1}{2}, y = \frac{-1}{2}$
4. $x = \frac{-1}{2}, y = \frac{1}{2}$

Question

Copy

Which of the following is a dimensional constant? **(1995)**

1. Relative density
2. Gravitational constant
3. Refractive index
4. Poisson's ratio.

Question

No Comm

The relation between $[E]$ and $[B]$ is

[JEE Adv, 2018]

1. $[E] = [B][L][T]$
2. $[E] = [B][L]^{-1}[T]$
3. $[E] = [B][L][T]^{-1}$
4. $[E] = [B][L]^{-1}[T]^{-1}$

$F = qE = qvB$ (hint)

$E = vB$

$E = B(LT^{-1})$

Question

The relation between $[\varepsilon_0]$ and $[\mu_0]$ is

[JEE Adv, 2018]

1. $[\mu_0] = [\varepsilon_0][L]^2[T]^{-2}$
2. $[\mu_0] = [\varepsilon_0][L]^{-2}[T]^2$
3. $[\mu_0] = [\varepsilon_0]^{-1}[L]^2[T]^{-2}$
4. ✓ $[\mu_0] = [\varepsilon_0]^{-1}[L]^{-2}[T]^2$

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \quad \text{(hint)}$$

$$c^2 = \frac{1}{\mu_0 \varepsilon_0}$$

$$\mu_0 = \frac{1}{\varepsilon_0 c^2}$$

$$\mu_0 = [\varepsilon_0^{-1}]\, L^{-2}\, T^2$$

Question

Q8

A physical quantity $\vec{S}$ is defined as $\vec{S} = (\vec{E} \times \vec{B})/\mu_0$, where $\vec{E}$ is electric field, $\vec{B}$ is magnetic field and μ_0 is the permeability of free space. The dimensions of $\vec{S}$ are the same as the dimensions of which of the following quantity (ies)? **(JEE Adv. 2021)**

1. $\dfrac{\text{Energy}}{\text{Charge} \times \text{Current}}$

2. $\dfrac{\text{Force}}{\text{Length} \times \text{Time}}$

3. $\dfrac{\text{Energy}}{\text{Volume}}$

4. $\dfrac{\text{Power}}{\text{Area}}$

$S = \dfrac{E \times B}{\mu_0}$

$F = BIl$

$F = \dfrac{\mu_0 I}{2\pi R} I l$

$F = \mu_0 I^2$ use

Question

Copy

Let us consider a system of units in which mass and angular momentum are dimensionless. If length has dimension of L, which of the following statement(s) is/are correct?

[JEE Adv, 2019]

1. The dimension of force is L^{-3}
2. The dimension of energy of L^{-2}
3. The dimension of power is L^{-5}
4. The dimension of linear momentum is L^{-1}

$L = r \times p$

$= m L T^{-2}$

$= L (L^2)^{-2}$

$= L \times L^{-4} = L^{-3}$

$= m L^2 T^{-2} = L^2 (L^2)^{-2}$

$= L^2 T^{-4} = T^{-2}$

$L = m^0 L^2 T^{-2} = m^0 L^0 T^0 = 1$

$L^2 T^{-1} = 1$

$L^2 = T^1$

Question

No cap

Expression for time in terms of G (universal gravitational constant), h (Planck constant) and c (speed of light) is proportional to

1. $\sqrt{\frac{hc^3}{G}}$
2. $\sqrt{\frac{c^3}{Gh}}$
3. $\sqrt{\frac{Gh}{c^5}}$
4. $\sqrt{\frac{Gh}{c^3}}$

Question

No

In terms of potential difference V, electric current I, permittivity ε_0, permeability μ_0 and speed of light c the dimensionally correct equations is/are **[JEE Adv, 2015]**

1. $\mu_0 I^2 = \varepsilon_0 V^2$
2. $\varepsilon_0 I = \mu_d V$
3. $I = E_0 c V$
4. $\mu_0 c I = \varepsilon_0 V$

Question

Young's modulus of elasticity Y is expressed in terms of three derived quantities, namely, the gravitational constant G. Planck's constant h and the speed of light c, as $Y = c^{\alpha}h^{\beta}G^{\gamma}$. Which of the following is the correct option? **[JEE Adv, 2023]**

1. $\alpha = 7, \beta = -1, \gamma = -2$
2. $\alpha = -7, \beta = -1, \gamma = -2$
3. $\alpha = 7, \beta = -1, \gamma = 2$
4. $\alpha = -7, \beta = 1, \gamma = -2$

Question

No (4)

Sometimes it is convenient to construct a system of units so that all quantities can be expressed in terms of only one physical quantity. In one such system, dimensions of different quantities are given in terms of a quantity x as follows: [position] = $[x^{\alpha}]$; speed = $[x^{\beta}]$; [acceleration] = $[x^{p}]$; [linear momentum] = $[x^{q}]$; [force = $[x^{r}]$. Then

[JEE Adv, 2020]

1. $\alpha + p = 2\beta$
2. $p + q - r = \beta$
3. $p - q + r = \alpha$
4. $p + q - r = \beta$

Question

No Lup

In a particular system of units, a physical quantity can be expressed in terms of the electric charge e, electron mass m_e, Planck's constant h, and coulomb's constant $k = \frac{1}{4\pi\varepsilon_0}$, where ε_0 is the permittivity of vacuum. In terms of these physical constants, the dimension of the magnetic field is $[B] = [e]^{\alpha}[m_e]^{\beta}[h]^{\gamma}[k]^{\delta}$. The value of $\alpha + \beta + \gamma + \delta$ is

(JEE Adv. 2022)

Question

Which of the following is a dimensional constant? **(1995)**

1. Relative density
2. Gravitational constant
3. Refractive index
4. Poisson's ratio.

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LIMITATION OF DIMENSIONAL ANALYSIS

(1) It is not use to derive dimensionless proportional constant.

(2) It can not derive dimensionless function, like $\sin\theta$, $\cos\theta$, $\tan\theta$ e^x etc.

(3) If physical quantity depends upon two P.Q. of same

(4) It can not derive formula which have 't' and '–' term

Ex : $S = ut + \frac{1}{2}at^2$

(5) We equate the power of M, L and T. So, it only work when quantity depends only on three physical quantity.

Ex : If force depends upon energy, velocity, time work.

Question

Which of the following equation can be derived dimensionally

1. $s = vt - \frac{1}{2}at^2$
2. $v^2 = u^2 - 2as$
3. $h = \frac{\omega^2 r^2}{2g}$
4. $v = \frac{d}{t}$

Question

Which of the following equation can not be derived dimensionally

1. $F = 6\pi r\eta$
2. $\theta = \omega t$
3. $\frac{d\theta}{dt} = \rho A v$
4. $P = \rho g h$

read & copy

Rule to find significant digit :

1. All non-zero are significant.
2. all zero between non-zero are significant.
3. Trailing zero without decimal point is not significant.
4. After decimal place all zero are significant
5. If number is less than one (1), all zero left of non-zero are not significant.
6. Exact number have infinite significant number.
7. In power form power is not significant.

Counting of Significant Digit

Counting of Significant Digit	
1.23.500 m	5
2.402 cm	3
3.5280 kg	3
4.6.780	4
5.0.034 gram	2
6.3.070 gram	4
7.1.608 × 10^{23}	4

Join must

H/W

→ Sanghorsh Assignment

→ Mahamanton of Vectg



Thank

You